INCH-POUND
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28 APRIL 2014

COMMERCIAL ITEM DESCRIPTION

TEST SET, SYNCHRO

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) Test Set Synchro provides a crystal controlled phase angle calibration standard. It produces two digitally synthesized, low distortion sine waves. Relative phase angles between two sine wave outputs may be varied by at least 0.002°over the range from ±999.999°. This instrument will provide and measure phase angle signals for development, repair and maintenance of general electronics, navigational aids, satellite and other communications equipment.

2. CLASSIFICATION. The Test Set Synchro covered by this (CID) shall be commercially available equipment that has factory installed options and/or may be modified to the extent necessary to meet the following description. The Test Set shall be Class 3, in accordance with MIL-PRF-28800.

3. SALIENT CHARACTERISTICS.

3.1 Item Description. The Test Set Synchro is a crystal controlled phase angle measurement calibration standard that produces two synthesized, low distortion sine waves.

3.2 Frequency Span. The requirement for the Test Set Synchro is to generate a signal with a span from 1 Hz to 200 kHz.

3.3 Frequency Resolution. The Test Set Synchro frequency resolution shall be a minimum of 0.002° up to 200 kHz.

Comments, suggestions, or questions on this document should be addressed to: AFLCMC/GUEEA, Robins AFB GA 31098-1813. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.,mil.

AMSC N/A FSC 6625

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3.4 Frequency Accuracy. The internal frequency accuracy shall be ≤ ±0.1 % at maximum frequency of 200 kHz.

3.5 Internal Time Base Stability. The internal reference frequency stability variation shall be ≤ $\pm 1X10^{-6}$ ppm for a 24 hour period.

3.6 Harmonic Distortion. The total harmonic distortion content of the Test Set Synchro shall be less than 0.5% average for phase measurements with ≤-56dB from reference from 1 Hz to 200 kHz (see table 1).

Table 1 Harmonic Distortion per Frequency Band

Harmonic Distortion % (dB)	Frequency
0.02%±0.05% (-70 dB)	1 Hz ~ 1000 Hz
0.07%±0.05% (-60 dB)	1000 Hz ~ 20 kHz
0.1%±0.05% (-55 dB)	20 kHz ~ 50 kHz
0.5%±0.25% (-48 dB)	50 kHz ~ 200 kHz

3.7 Synchro Output Level. The Test Set Synchro shall have output amplitude levels and resolutions as shown in (see table 2). Minimum output current (I_{out}) shall be 15 Ma.

Table 2 Output Amplitude Range

Amplitude Range	Resolution (≤)
100mV~500mV	1mV
.5V~8V	2mV
8V~100V	33.33mV
100V~120V	100mV

3.8 Power Output Accuracy. The signal level shall be fully adjustable with accuracies within the values (see table 3).

Table 3 Amplitude Accuracy

Amplitude Accuracy (≤)	Frequency Band
±5 mV (±0.5%)	1Hz to 50 kHz
±25 mV (±1%)	50 kHz to 100 kHz
±50 mV (±5%)	100 kHz to 200 kHz

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3.9 Phase Angle Measurements. The relative phase angle between two waveform outputs may be measured with a minimum difference value of 0.002° over a range from ±999.999° (See table 4).

Table 4 Phase Angle Measurements

Phase Angle Measurements (Range from ±999.999°, 1 Hz to 200 kHz, Resolution ±0.002°)				
Voltage	**Frequency and Angle Tolerance**			
	1 Hz - 1 kHz	1 kHz - 6.25 kHz	6.25kHz - 50 kHz	50 kHz - 200 kHz
5V	±5 m°	±5 m°	±10 m°	±40 m°
5V~100V	±5 m°	±10 m°	±15 m°	±40 m°
100V~120V	±10 m°	±20 m°	±30 m°	±80 m°

3.10 Triggering and Remote Control. The Test Set Synchro shall have triggering (enable/disable), parameter value entry capability through the GPIB port by means of the IEEE 448-1 control and communications protocol under the IEC-60488-1, Standard Digital Interface for Programmable Instrumentation.

3.11 Zero Reference Measurement. The Test Set Synchro shall be capable of performing an "Auto Zero" reset value upon power up or on demand by the user.

3.12 Design and Construction. The Test Set Synchro shall be used in both bench top and rack mount environments. The Test Set Synchro is a commercial off the shelf item subject to commercial products regulatory standards. The Test Set Synchro shall be designed and constructed in accordance with all applicable European Union (EU) requirements with the "CE" marking affixed, International Organization for Standardization, the American Society for Quality ASQ/ISO/IEC 17025 standard, the International Telecommunication Union (ITU), Underwriters Laboratories (UL) and the American National Standards Institute (ANSI). It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service

3.13 System Safety. The Test Set Synchro shall comply with all sections of the Underwriters Laboratories (UL) standard 1449 for electrical safety. The system shall comply with MIL-STD-1472 section 5.9.11.3.3 for weight & size restrictions for single person lift & handling.

3.14 Sound Levels. The Test Set Synchro/Resolver Standard acoustic noise shall conform to sound level requirements ISO 7779 and ISO 8253-2.

3.15 Electrostatic Discharge (ESD). The design of the Test Set Synchro shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking in accordance with MIL-STD-1686 section 5.2.2.3.

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3.16 Electro Magnetic Interference (EMI). The Test Set Synchro EMI emissions shall be in accordance with the following radiated emission and susceptibility requirements of MIL-PRF-28800, MIL-STD-461: RE102 and RS103 & IEC-1000-4 thru IEC-1000-6. Emitted radiation absorption and radiation limits FCC Part 15.109, Part 15.209 and IEC-61000 sub volume 4-3.

3.17 Service Life. The design of the Test Set Synchro shall be such that under normal use and operation the service repair can be performed according to MIL-PRF-28800 sections 3.14.2 & 3.14.3.

3.18 Port Damage. The input and output port(s) shall have a protection method that can be achieved by either internal or external fusing

3.19 Connectors. The system shall have a BNC type connectors outputs. A General Purpose Interface Bus (GPIB) interface with IEEE-488-1 connector is required for compatibility and control functions. The connectors shall comply with MIL-PRF-39012 section 3 for requirements.

3.20 Operating Temperature Range. The Test Set Synchro shall meet the specifications listed herein with full capabilities within the temperature range of 50°F to 122°F (10°C to 50°C) after tests in MIL-PRF-28800 section 4.5.5.1.1 have been performed for class 3 equipment classification (Per table 11 on page 41).

3.21 Storage Temperature Range. The Test Set Synchro shall meet and/or exceed manufacturers specifications for stored equipment after tests of MIL-PRF-28800 section 4.5.5.1.1 temperature, humidity & storage have been performed for instruments in the range of class 3 equipment classification (Per table 11 on page 41).

3.22 Altitude Requirements. The Test Set Synchro shall meet its operating performance and accuracy while operating at an altitude of 8,000 ft. (2,438 meters). The equipment shall meet its performance and accuracy requirements, for operation and non-operating storage, as specified in MIL-PRF-28800 section 3.8.3 and section 4.5.5.2 respectively at this altitude.

3.23 Heat Dissipation and Ventilation. Ventilation shall be sufficient to allow for prevention, buildup and retention of moisture, heat dissipation either passive or active shall be suitable in order to meet MIL-PRF-28800 section 4.5.5.1.1 for class 3 equipment classification (Per table 11 on page 41).

3.24 Calibration & Maintenance Adjustments. No special tools, software, technical manuals or documentation shall be required for the alignment, adjustment, repair and calibration of the Test Set Synchro except those provided with the test set.

3.25 Calibration Period. The design of the Test Set Synchro shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the test set case or modules. The calibration interval shall be based on a period of one calendar year or an operating time of 2,080 hours, whichever occurs first. At the end of this interval 85% of the units shall remain within calibration tolerances in accordance with MIL-PRF-28800 section 3.12.

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3.26 Calibration & Maintenance Adjustments. No special tools or software shall be required for alignment or adjustment except those furnished with the Test Set Synchro test set. A software license shall be issued for each physical instrument when the system is designed for computer based calibration. Any special equipment, fixtures, tools, standards, etc., required for calibration and not common to a secondary calibration laboratory shall be supplied with the Test Set Synchro. Any unique accessories including software required for calibration of this item shall be furnished by the manufacturer. Units that fail calibration and are recommended to be aligned or adjusted, either manually or using an automated alignment/adjustment software, the special hardware, manual instructions, software, software instructions and software licenses (where required by the manufacturer, and shall be valid for the life of the instrument) needed to perform the alignment/adjustment process shall be included with the Test Set Synchro. A list of peripheral support TMDE equipment required to accomplish the calibration/alignment/adjustments of the unit shall be provided with the Test Set Synchro. Any unique accessories including software required for calibration of this item shall be furnished by the manufacturer. All firmware or software provided under this purchase description shall be the same version, revision and revision date.

3.27 Reliability. The design of the Test Set Synchro shall be such that under normal use and operation the mean time before failure (MTBF) exceeds 2,080 hours of operation with a statistical certainty equal or greater than 95% in accordance with MIL-PRF-28800 section 3.13. The manufacturer shall provide a summary of maintenance or warranty records to establish the validity of the statistical assertion in accordance with MIL-PRF-28800 section 3.13. The manufacturer shall be able to reasonably establish the claim that the offered equipment is capable of the required level of reliability by meeting and/or exceeding MIL-PRF-28800 sections 3.13 and section 4.5.8.1.

3.28 Built in Test Equipment and Diagnostics. The Test Set Synchro shall be equipped with built in test and diagnostic capability that at “Power On or “Cold Start” performs a system test evaluating all functional characteristic of the instrument in accordance with section 6.5.2.6 of MIL-PRF-28800.

3.29 Maintainability. The Test Set Synchro shall meet the human engineering factors of ergonomics for operational & maintainability criteria, providing access to controls, internal components and adjustments in accordance with MIL-STD-1472 paragraphs 5.9 through 5.9.18 and MIL-PRF-28800 section 6.6.2.4.

3.30 Preventive Maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800 section 3.14.2.

3.31 Corrective Maintenance. Test Set Synchro shall be designed to meet reparability criterion in accordance with section 3.14.1 of MIL-PRF-28800.

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3.32 Electrical Power Sources & Connections. The Test Set Synchro equipment shall operate from nominal commercial and military power sources with a nominal voltage of 120 V_{RMS} Single Phase AC (±10%) at line frequencies of 50Hz or 60Hz in accordance with MIL-PRF-28800 paragraph 3.1.5.2 and 3.1.5.2a.

3.33 Electrical Surge. The Test Set Synchro shall be able to provide surge protection that meets section 38 of the UL 1449 standard for equipment and personnel protection.

3.34 Noise Suppression. The Test Set Synchro shall be able to operate with electrical noises and harmonic content present on the alternating current power source with frequencies in the 30Hz to 50 kHz range and levels varying from line level of 41.6 dB ±1 dB (108 VAC to 132 VAC) measured at the power input in accordance with IEEE-519 standard.

3.35 Dimensions. The Test Set Synchro shall comply with width and depth of EIA/ECA-310-E standard equipment rack configuration. The height shall be compatible with 4 standard size height 4U. One unit or "U" is defined by EIA/ECA-310-E to be 1.75 inches (44.45 mm).

3.36 Weight. The total weight of the Test Set Synchro, excluding accessories and manuals, shall comply with MIL-STD-1472 section 5.9.11.3.3 for weight & size restrictions for single person lift & handling and shall not exceed a one-person lift of 35 pounds, (15.6 kilograms).

3.37 Rack Mount. The Test Set Synchro shall be capable of being rack-mounted in a standard EIA/ECA-310-E equipment rack and if required include a rack-mount conversion kit that meets the requirements of section 3.6.4.14.1 of MIL-PRF-28800.

3.38 Markings. All warning decals and instrument category rating must conform to safety standards and be visible by user in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible. Markings, Information/Caution must be Lusterless Black, and Markings, Warning/Danger shall be Lusterless Red.

4. REGULATORY REQUIREMENTS.

4.1 Recycled recovered materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price.

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The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

5.1 Verification. Verification of the delivered SA will be performed at the delivery point by the end user

5.2 First Production Inspection. The first production item shall be subjected to the analyses, demonstrations, examinations, and tests conducted by the Air Force Metrology and Calibration Division. The supplier shall provide or arrange the delivery of the equipment in its final delivery configuration for test and evaluation

6. PACKAGING. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. The inspection of the preservation, packaging, and packing shall be in accordance with the requirements of ASTM D4169, assurance level I. The acceptance criteria are that the equipment is damage-free and that the package is intact. Markings shall be inspected to assure that the required information is present and that the marking is legible, nonfading, and durable. When actual packaging of the materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

7. NOTES.

7.1 Source of documents.

7.1.1 Military Specifications, Standards and Handbooks referenced herein may be obtained at https://assist.dla.,mil or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg. 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 EIA/ECA documents can be obtained online at http://www.eciaonline.org/default.aspx .

7.1.3 ISO documents can be obtained online at http://www.iso.org or from the International Organization for Standardization, ISO Central Secretariat, 1, ch de la Voie-Creuse, CP 56-CH-1211 Geneva 20, Switzerland.

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7.1.4 FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh PA 15250-7954. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/.

7.1.5 ANSI/ASME documents may be obtained online at http://www.asme.org/ or from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

7.1.6 IEEE copies can be obtained online at www.**ieee**.org .

7.1.7 Underwriters Laboratories (UL) copies may be obtained online at http://ulstandardsinfonet.ul.com or from: COMM 2000, 1414 Brook Drive, Downers Grove, IL 60515-5000.

7.1.8 ASTM documents. Application for copies should be addressed to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org .

7.2 Key words.

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